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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Dirty Cafe Sales Data Cleaning and Preparation Using Power Query

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Experiment No.: 4

Dirty Cafe Sales Data Cleaning and Preparation Using Power Query

GITHUB Link: <https://github.com/pragathidevanga/Business-Analytics>

Problem Statement

A company has received customer transaction data containing missing values, errors, duplicate records, inconsistent text entries, and incorrect or unsuitable data types. Such raw data cannot be directly used for reliable business analysis because inaccurate or inconsistent records may affect the results.

The objective of this experiment is to clean and prepare the Dirty Cafe Sales Dataset using Power Query in Power BI. The cleaning process involves identifying data quality issues and applying suitable transformation operations such as replacing errors and unknown values, correcting data types, creating required numerical fields, removing unnecessary columns, standardizing text values, filtering invalid records, and removing duplicate transactions.

The final objective is to obtain a clean, consistent, and analysis-ready dataset that can be used for further business analytics and visualization.

Methodology / Steps Followed

The complete data preparation process was carried out using Power Query Editor in Power BI. The major steps followed are:

1. Data Collection
2. Data Import
3. Data Cleaning and Transformation
4. Final Data Validation and Preparation
5. Output and Verification

Step 1 – Data Collection

Dataset Description

The dataset used for this experiment is the **Dirty Cafe Sales Dataset**, obtained from **Kaggle**. It contains transaction-level information related to cafe sales.

The dataset includes information about the transaction, item, quantity, price, total amount spent, payment method, location, and transaction date.

Since the dataset is intentionally dirty, it contains several data quality problems such as errors, unknown values, inconsistent text entries, duplicate records, and values that require validation.

The dataset was selected because it provides a practical example of how real-world business data may contain inconsistencies and why data cleaning is an important stage before performing analysis.

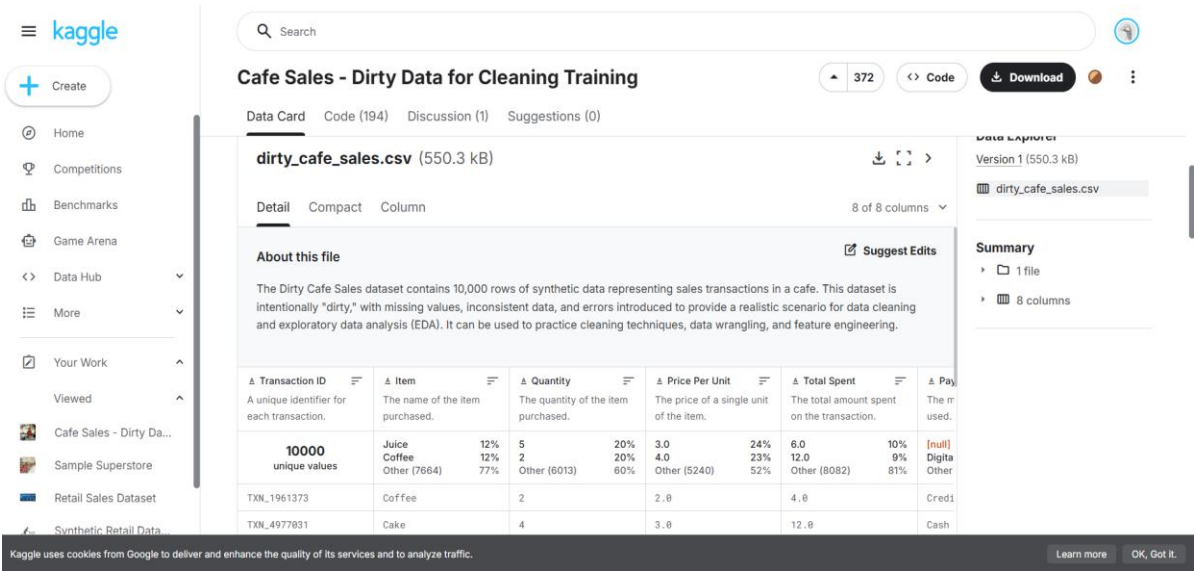


Fig. 1: Kaggle Dirty Cafe Sales Dataset

Dataset Information

Attribute	Details
Dataset Name	Dirty Cafe Sales Dataset
Source	Kaggle
Data Type	Business Transaction Data
File Type	Flat File / CSV
Tool Used	Power BI – Power Query
Number of Columns	8
Main Purpose	Data Cleaning and Preparation
Final Use	Business Analytics

Dataset Columns

The original dataset contains the following eight columns:

Column Name	Description
Transaction ID	Unique identifier associated with each transaction
Item	Name of the cafe item purchased
Quantity	Number of units purchased
Price Per Unit	Price of one unit of the purchased item
Total Spent	Total amount spent in the transaction
Payment Method	Method used by the customer to make payment
Location	Location associated with the transaction
Transaction Date	Date on which the transaction occurred

Data Quality Issues Identified

Before cleaning the dataset, different types of data quality problems were identified. These included:

- Missing or invalid values
- Error values in numerical columns
- Unknown values
- Incorrect data types
- Inconsistent item names
- Inconsistent payment method entries
- Inconsistent location values
- Invalid or unwanted dates
- Duplicate transaction records
- Extra spaces in text values
- Unclean characters in text fields
- Invalid quantity values

- Invalid price-per-unit values

These issues were handled systematically using Power Query transformations.

Step 2: Data Import

The Dirty Cafe Sales Dataset was imported into Microsoft Power BI for cleaning and preparation.

The following procedure was followed:

- Opened Power BI Desktop.
- Used the Get Data option to import the dataset.
- Selected the Dirty Cafe Sales dataset.
- Selected Transform Data to open the Power Query Editor.
- Examined the imported columns and their initial values.
- Prepared the dataset for further cleaning and transformation.

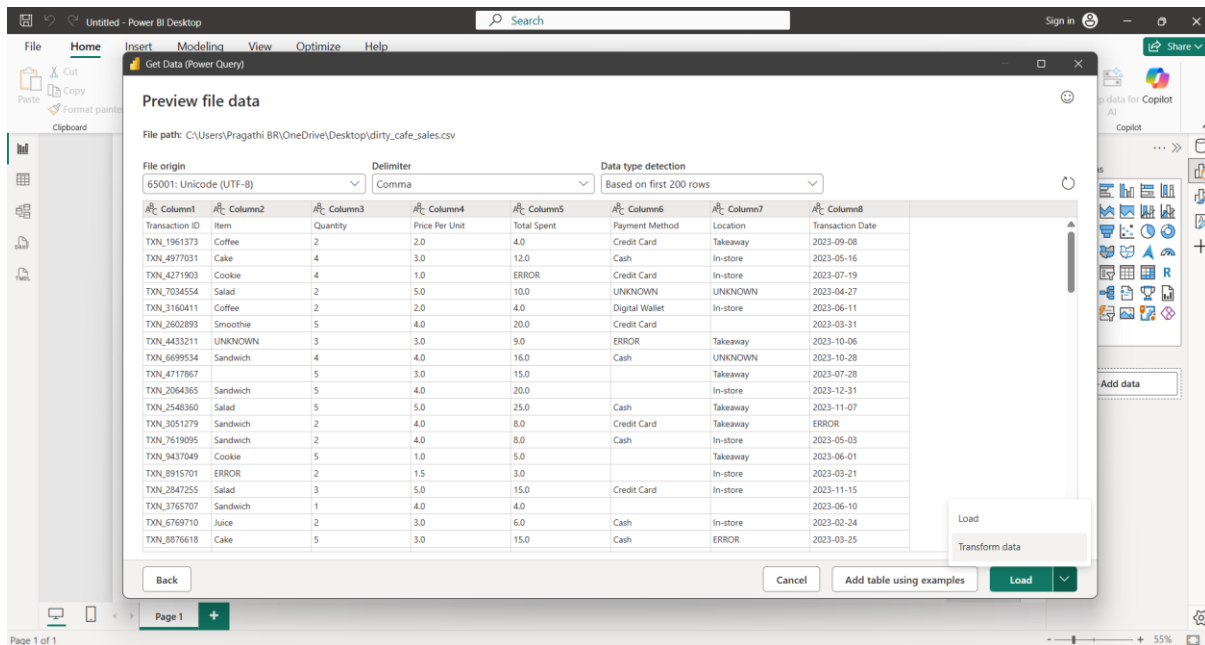


Fig. 2: Dataset Imported into Power Query

Step 3 – Data Cleaning and Transformation

The Dirty Cafe Sales Dataset was cleaned and prepared using Power Query in Power BI. The dataset contained different types of data-quality issues, including errors, unknown values, inconsistent entries, duplicate records, unnecessary columns, and values that required appropriate data types. Power Query was used to systematically identify and transform these issues.

The cleaning process was performed through a sequence of Applied Steps. Each step had a specific purpose in preparing the dataset for further business analysis. The transformations also created a clear history of the changes made to the original dataset, making the cleaning process easy to track and reproduce.

1. Source

The Source step is the initial stage of the Power Query process. It represents the connection between Power Query and the Dirty Cafe Sales Dataset obtained from Kaggle. The complete raw dataset was loaded into Power Query so that its columns and records could be inspected before applying any transformations.

At this stage, the original information was retained as the input data. The dataset contained transaction-related information including Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date. This source data formed the foundation for all the subsequent cleaning operations.

The Source step is important because every transformation performed later in Power Query is based on the data loaded at this stage.

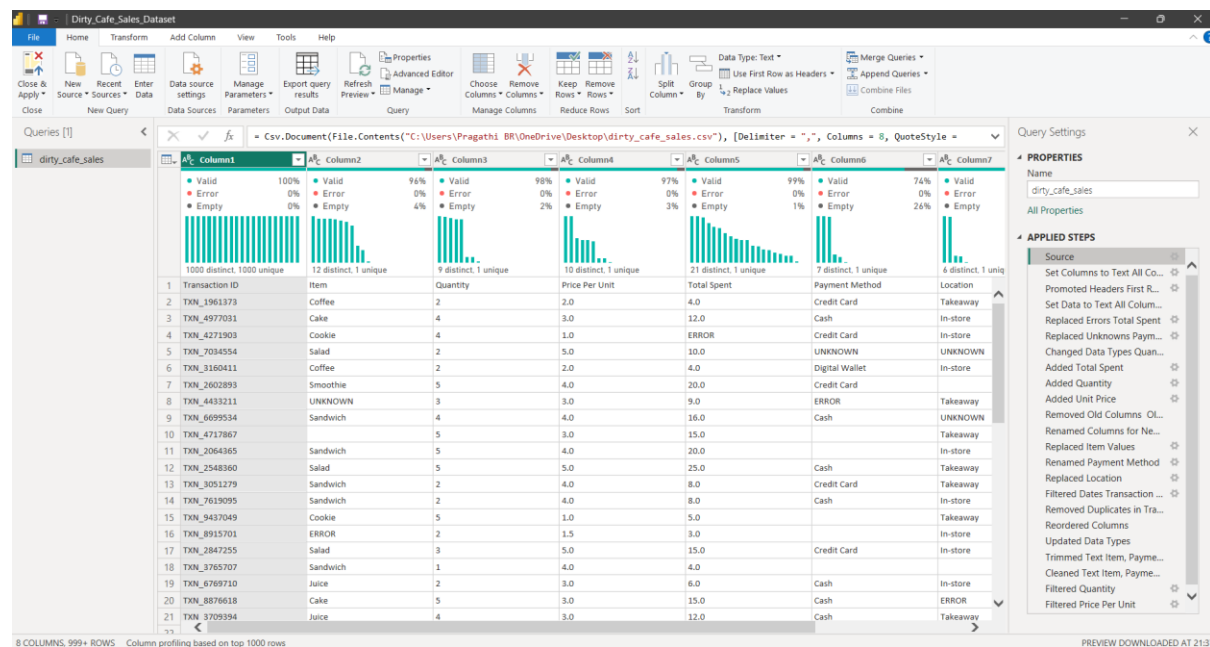


Fig. 3: Source step in Power Query

2. Set Columns to Text All Columns

The Set Columns to Text All Columns step was used during the initial preparation of the dataset. The columns were set to text format so that the raw values could be handled consistently before their final data types were assigned.

This was useful because the dataset contained different types of entries, including valid values, errors, unknown values, and inconsistent information. Treating the columns as text initially allowed these values to be inspected and cleaned without unwanted automatic type-conversion problems.

The step therefore provided a controlled starting point for the subsequent data-cleaning operations.

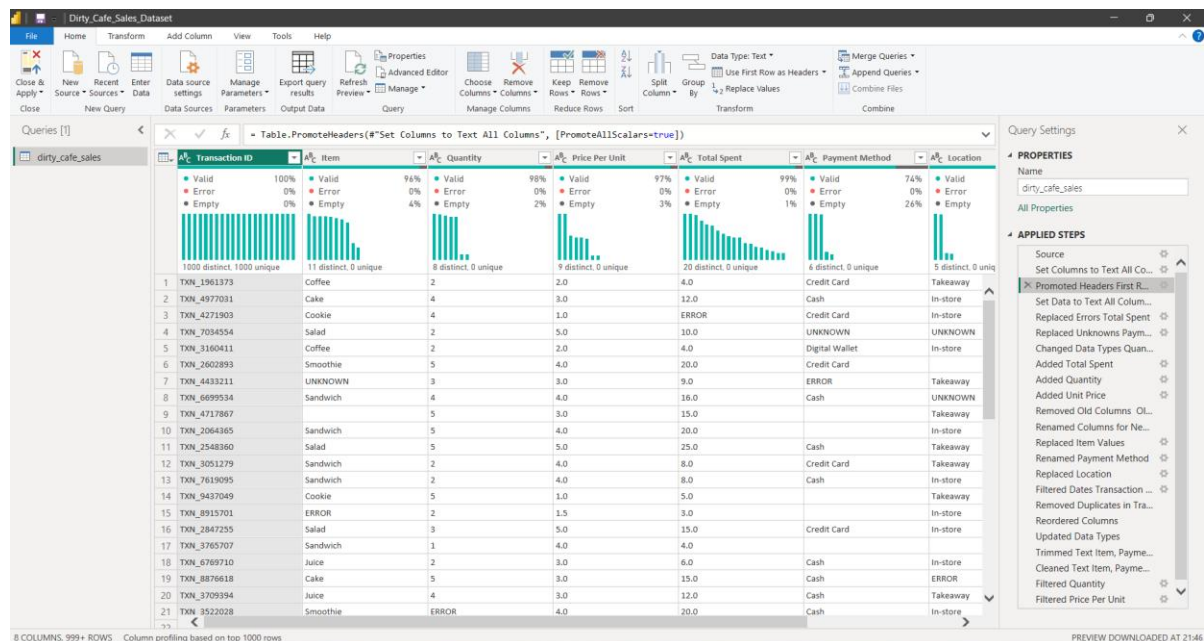


Fig. 4: Set Columns to Text All Columns

3. Promoted Headers First Row

The Promoted Headers First Row step was used to convert the first row of the imported dataset into the column-header row.

When a flat-file dataset is initially loaded, Power Query may treat the first row as ordinary data. Promoting the first row ensures that the actual field names are recognized as headers. This makes the dataset easier to understand and allows the columns to be referred to using meaningful names.

After this operation, the transaction fields could be clearly identified, making the remaining transformations easier to perform.

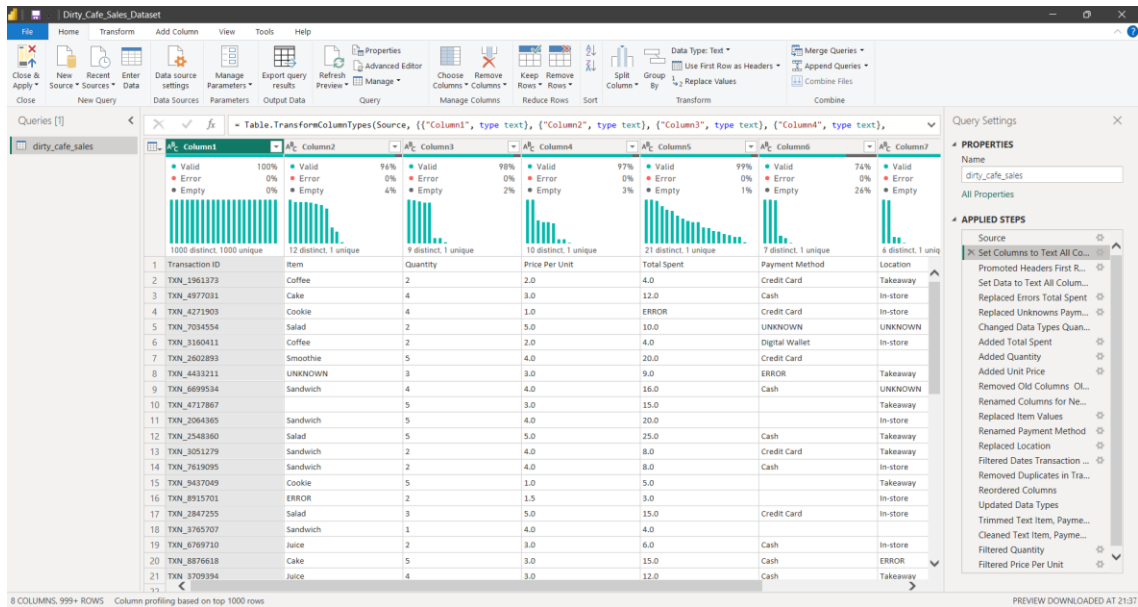


Fig. 5: Promoted Headers First Row

4. Set Data to Text All Columns

The Set Data to Text All Columns step was used to prepare the dataset for the next stages of cleaning. The data fields were maintained in text format while the problematic values were being handled.

This approach helped prevent Power Query from incorrectly interpreting mixed or dirty values before the cleaning process was completed. It was particularly useful for a dataset containing errors and inconsistent entries.

The appropriate final data types were applied later after the necessary transformations had been performed.

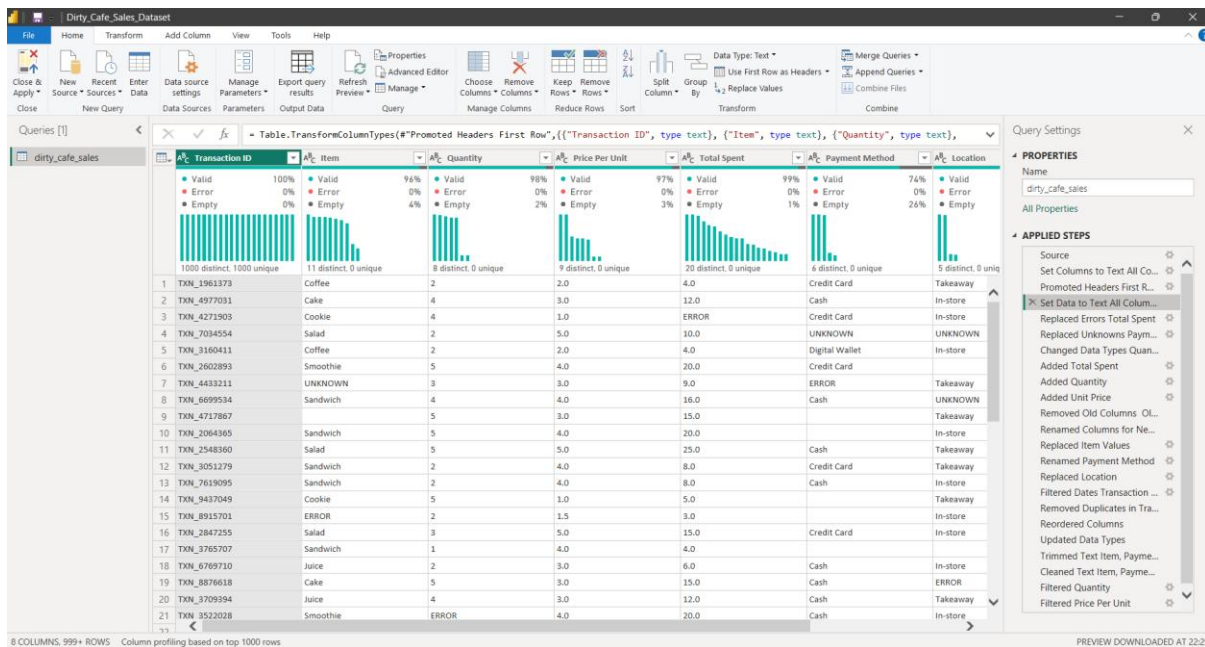


Fig. 6: Set Data to Text All Columns

5. Replaced Errors Total Spent

The Replaced Errors Total Spent step was specifically applied to the Total Spent field. The dataset contained error values in this field, which needed to be handled before the column could be used reliably.

Error values can create problems during calculations and may prevent Power BI from correctly processing a numerical field. Therefore, the errors were addressed during the cleaning stage rather than allowing them to remain in the final dataset.

This step improved the quality of the Total Spent field and prepared it for later data-type and validation operations.

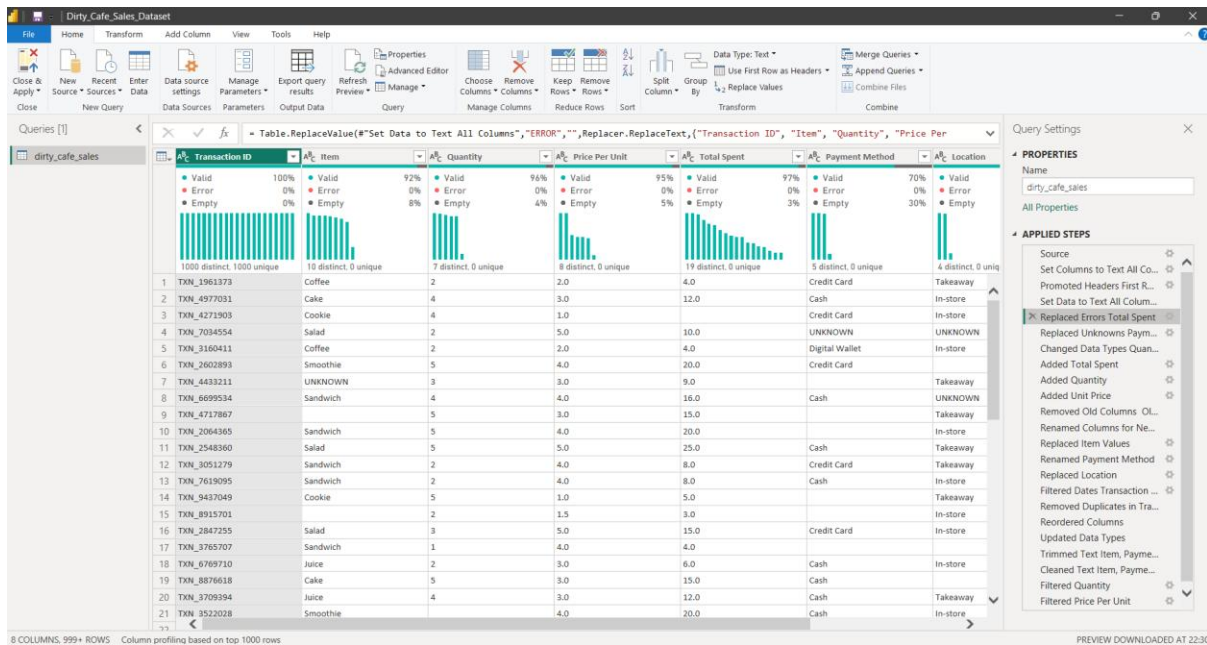


Fig. 7: Replaced Errors Total Spent

6.Replaced Unknowns Payment Method

The Replaced Unknowns Payment Method step was used to handle unknown entries in the Payment Method field.

Payment Method is a categorical field that identifies how customers completed their transactions. Unknown or inconsistent entries can make category-based analysis difficult because the same type of information may be represented differently.

The replacement operation helped standardize the payment-related information and improved the overall consistency of this field. This makes the data more suitable for future grouping and comparison.

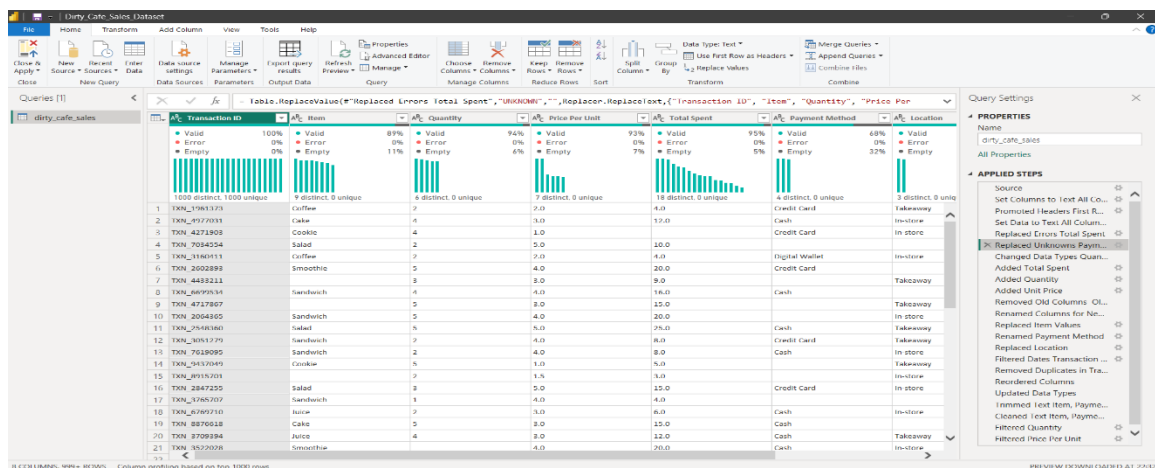


Fig. 8: Replaced Unknowns Payment Method

7.Changed Data Types Quantity

The Changed Data Types Quantity step was used to assign a suitable data type to the Quantity field.

Quantity represents the number of units purchased in a transaction and therefore needs to be handled as numerical information. Applying the appropriate data type allows Power BI to perform numerical operations, filtering, comparison, and validation correctly.

This step ensured that the Quantity field was prepared for the later filtering and data-cleaning operations.

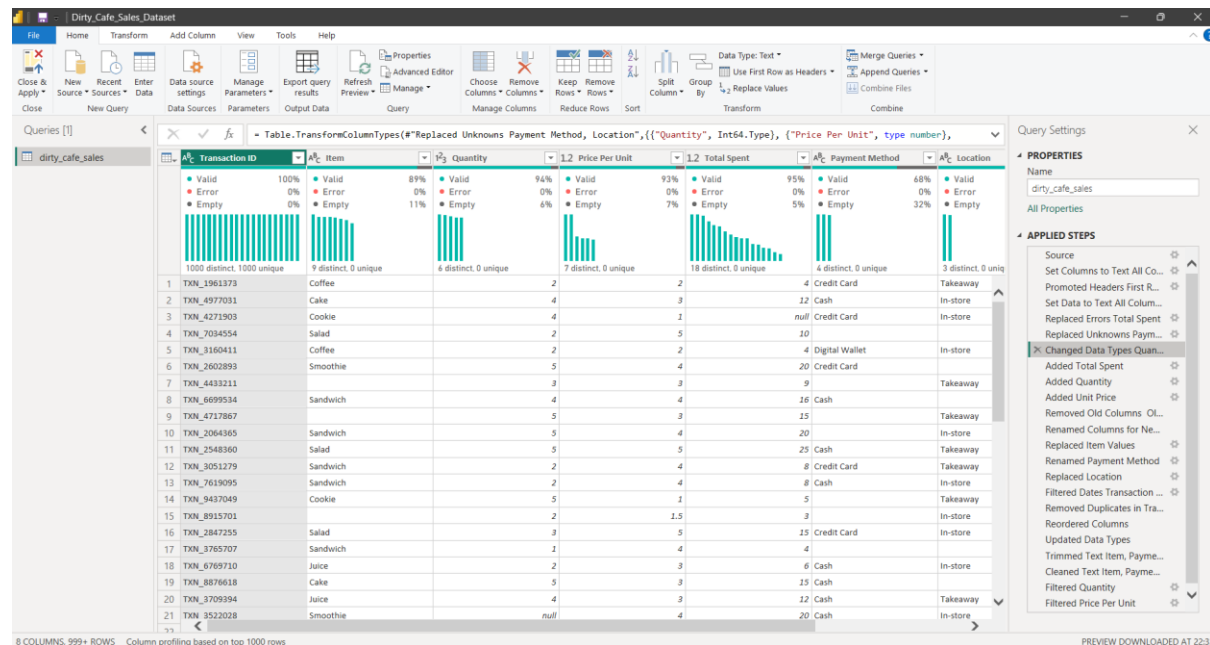


Fig. 9: Changed Data Types Quantity

8. Added Total Spent

The Added Total Spent step was used to prepare the required Total Spent information in the transformed dataset.

Total Spent is an important business field because it represents the transaction amount associated with a purchase. It is useful for understanding the monetary value of transactions and can later support business calculations and analysis.

This step formed part of the preparation of the numerical fields before the older or unnecessary versions of these fields were removed.

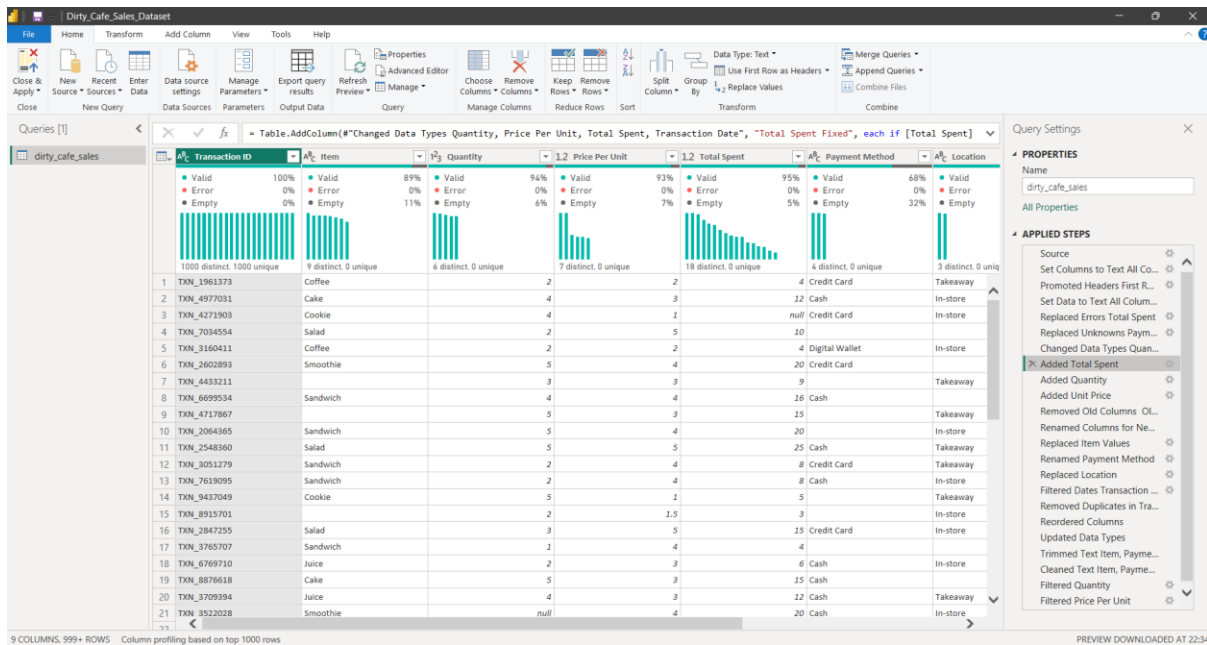


Fig. 10: Added Total Spent

9. Added Quantity

The Added Quantity step was used to prepare the required Quantity field in the dataset.

Quantity provides information about the number of units purchased during each transaction. Maintaining this information correctly is important because it can be used to understand purchase volumes and perform numerical analysis.

The step ensured that the required Quantity information was available before the dataset proceeded through the remaining cleaning operations.

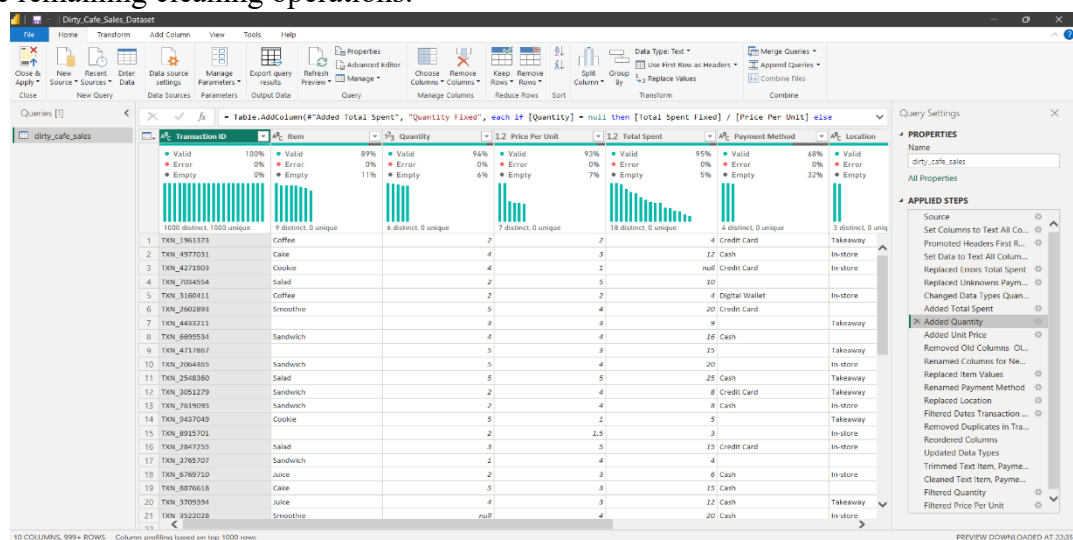


Fig. 11: Added Quantity

10. Added Unit Price

The Added Unit Price step was used to prepare the unit-price information in the dataset.

Unit Price represents the price associated with one unit of the purchased item. It is an important numerical field because it contributes to understanding the pricing information associated with each transaction.

Preparing this field at this stage allowed the required numerical information to be retained before unnecessary or older columns were removed.

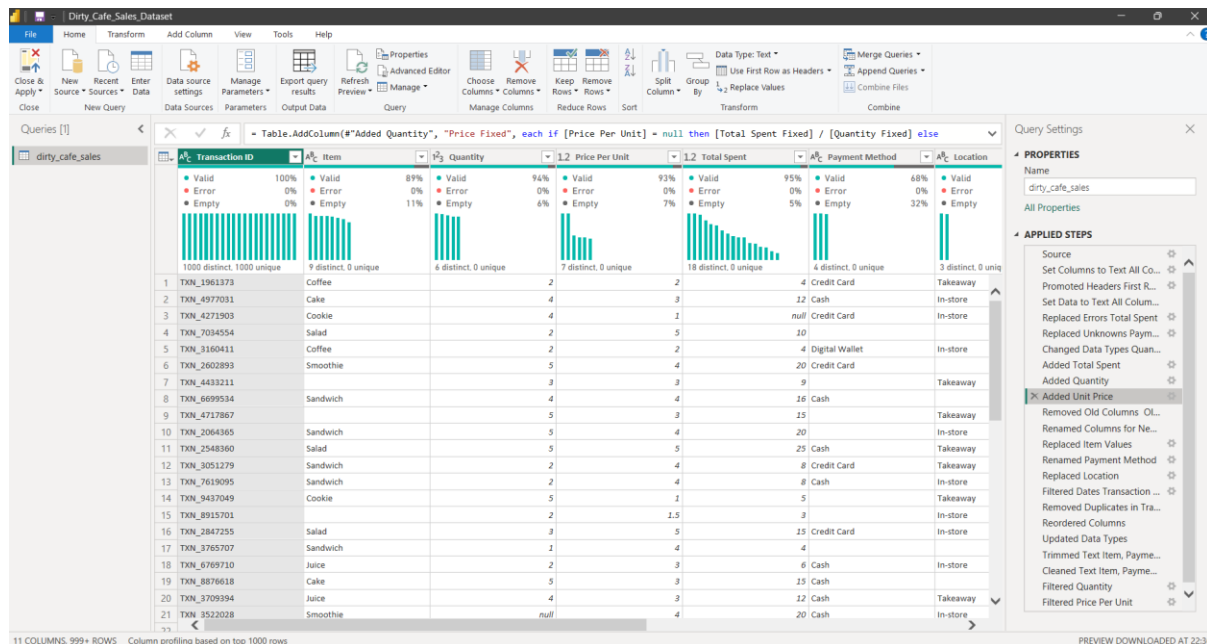


Fig. 12: Added Unit Price

11. Removed Old Columns

The Removed Old Columns step was used to remove older or unnecessary column versions that were no longer required after the new fields had been prepared.

Keeping duplicate or outdated versions of the same information can make a dataset difficult to understand and may create unnecessary redundancy. Removing these fields simplified the dataset and retained the information required for the remaining transformations.

This step therefore improved the structure and organization of the dataset.

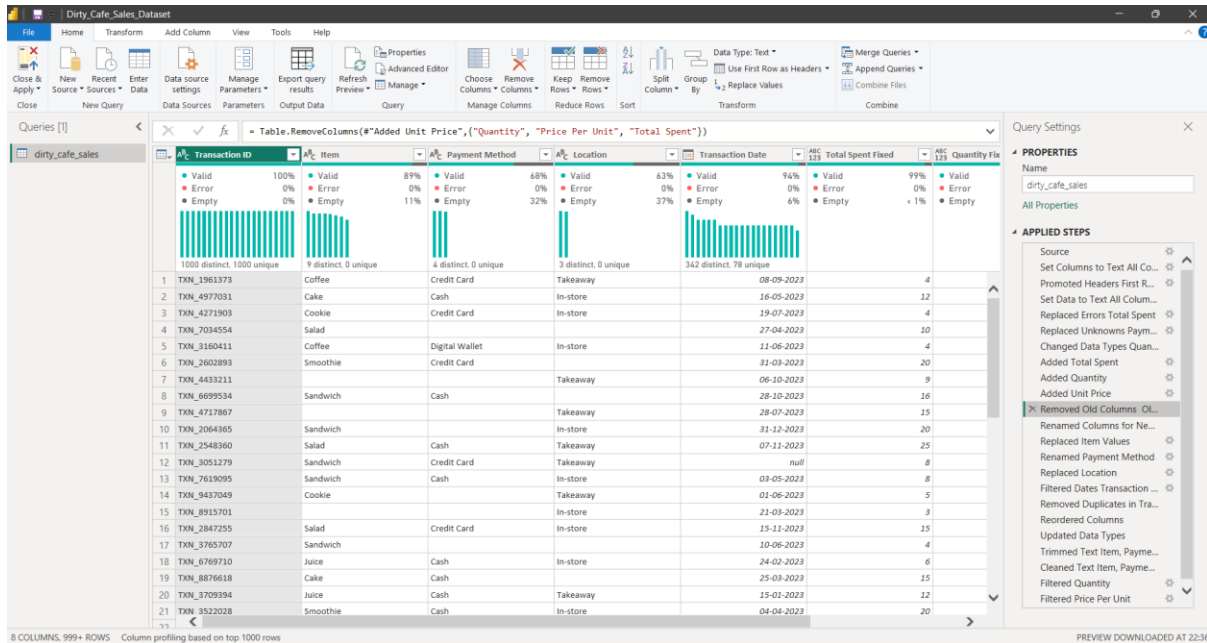


Fig. 13: Removed Old Columns

12. Renamed Columns for New calculated columns

The Renamed Columns for New calculated columns step was used to rename the required columns into clearer and more meaningful names.

Clear column names are important in business datasets because they allow users to quickly understand what each field represents. Renaming also improves the readability of the dataset and makes it easier to work with the fields during future analysis.

This step helped create a more understandable and professionally structured dataset.

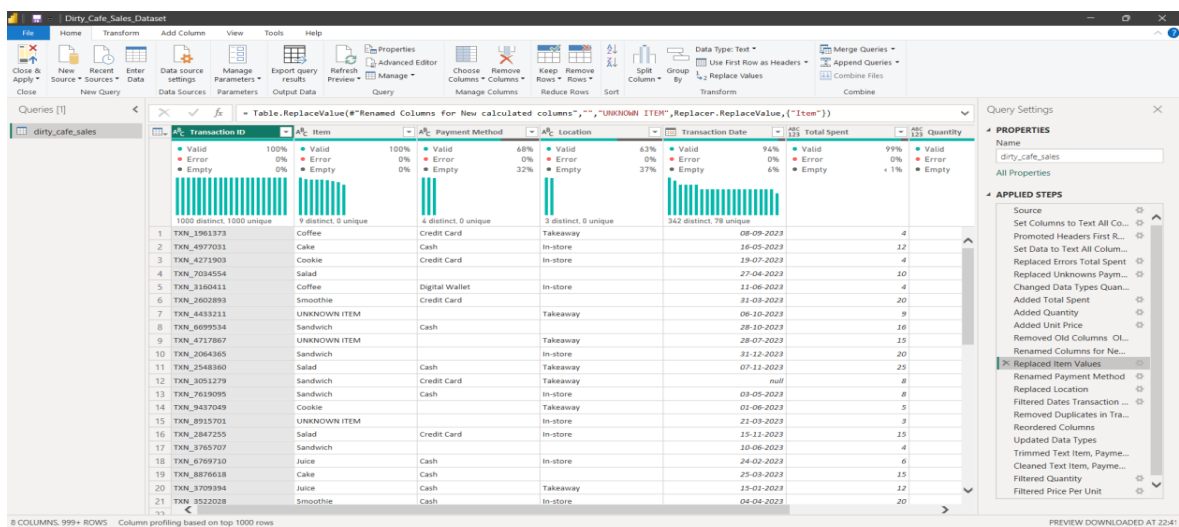


Fig. 14: Renamed Columns for New calculated columns

13. Replaced Item Values

The Replaced Item Values step was applied to the Item column to handle inconsistent item entries.

In a business transaction dataset, item names need to be represented consistently so that the same item is not incorrectly treated as multiple categories. Replacing the required values helped standardize the information contained in the Item field.

This improved the quality of the product-related data and prepared it for reliable categorical analysis.

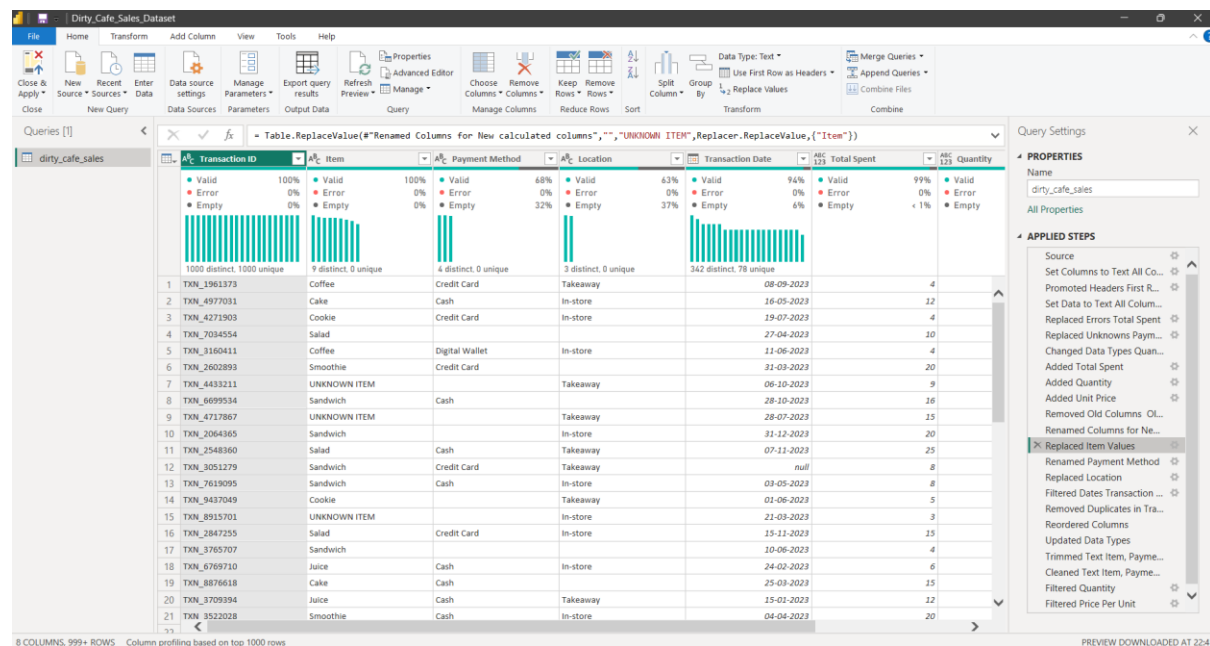


Fig. 15: Replaced Item Values

14. Renamed Payment Method

The Renamed Payment Method step was used to provide a clear and understandable name for the Payment Method field.

A meaningful field name improves the readability of the dataset and makes it easier for a user to identify the purpose of the column. This is particularly useful when working with business data containing multiple transaction-related fields.

The operation therefore improved the organization and presentation of the dataset.

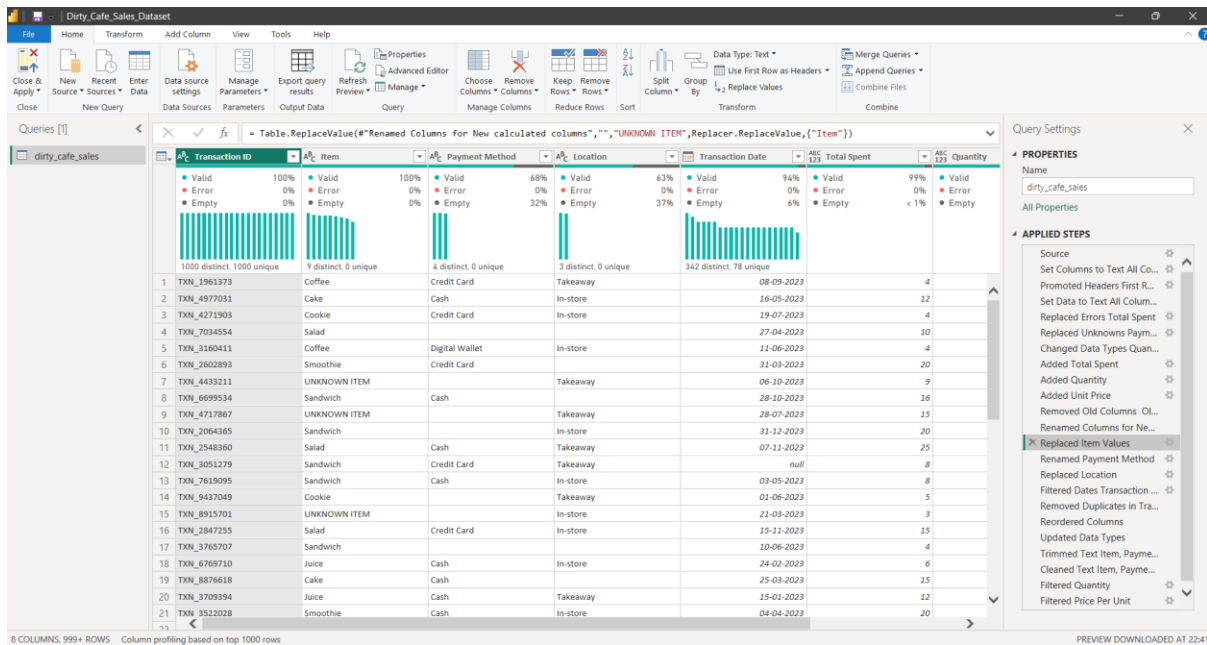


Fig. 16: Renamed Payment Method

15. Replaced Location

The Replaced Location step was applied to the Location field to handle inconsistent location entries.

Location information is useful for comparing transactions across different places. If similar locations are represented using inconsistent values, the results of location-based analysis may become inaccurate.

The replacement operation helped standardize the required location information and improved the consistency of the dataset.

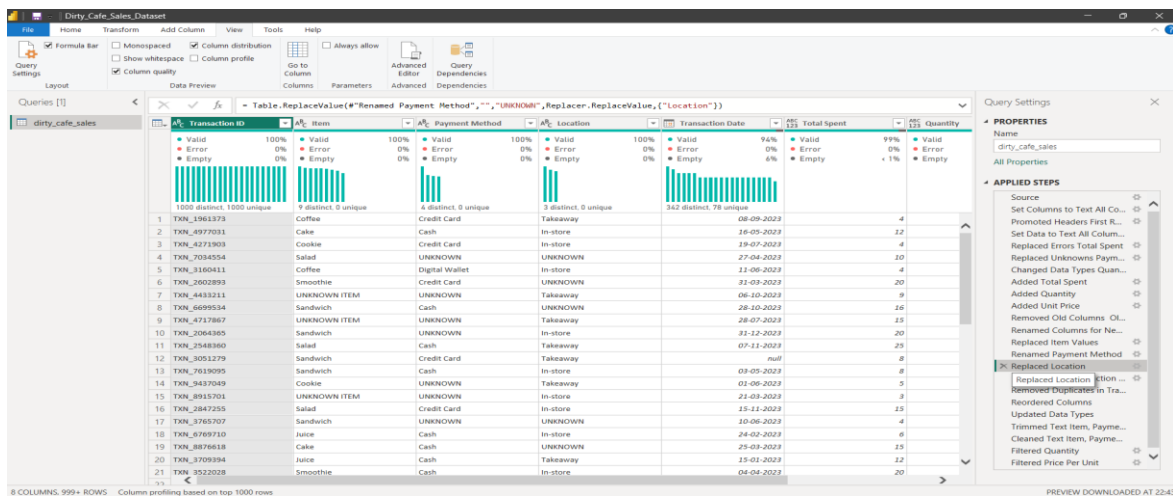


Fig. 17: Replaced Location

16. Filtered Dates Transaction Date

The Filtered Dates Transaction Date step was applied to the Transaction Date field.

Transaction Date is an important field because it identifies when each purchase took place. The filtering operation was used to retain the required date records according to the conditions applied during the cleaning process.

This helped ensure that unsuitable date records were not carried forward into the final prepared dataset and that the remaining transaction information was suitable for further analysis.

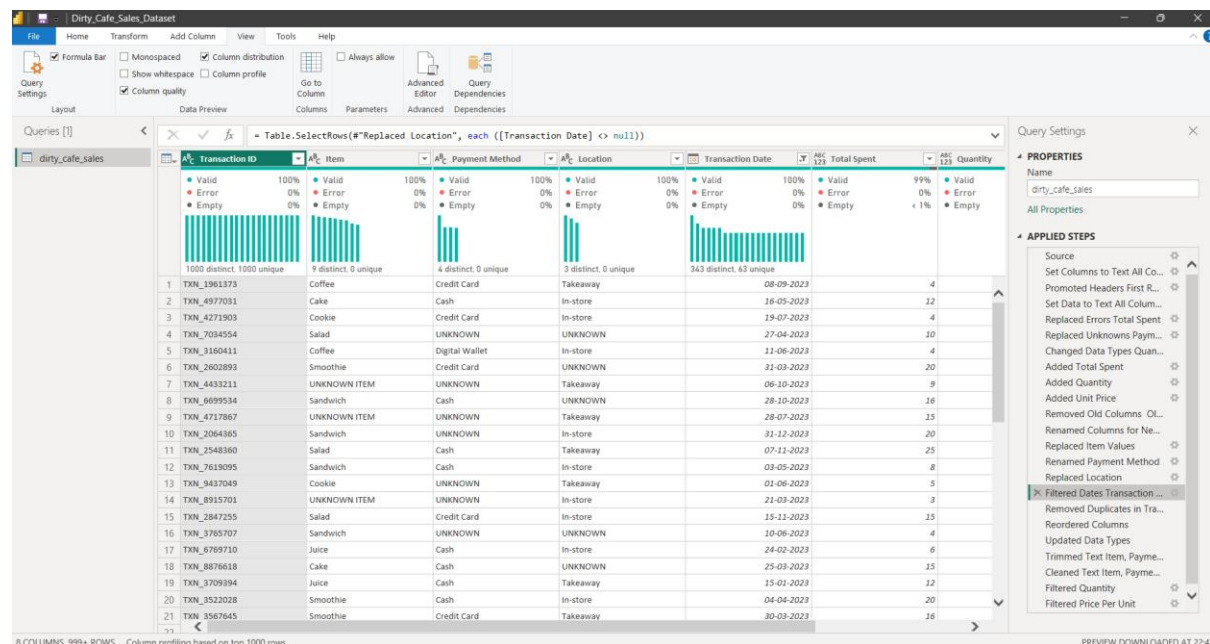


Fig. 18: Filtered Dates Transaction Date

17. Removed Duplicates in Transaction ID

The Removed Duplicates in Transaction ID step was used to remove duplicate transaction records.

Duplicate records can cause inaccurate business results because the same transaction may be counted more than once. Using the transaction information for duplicate removal helped improve the uniqueness and reliability of the dataset.

This was an important data-quality operation because a clean transaction dataset should avoid repeated records wherever duplicate entries are identified.

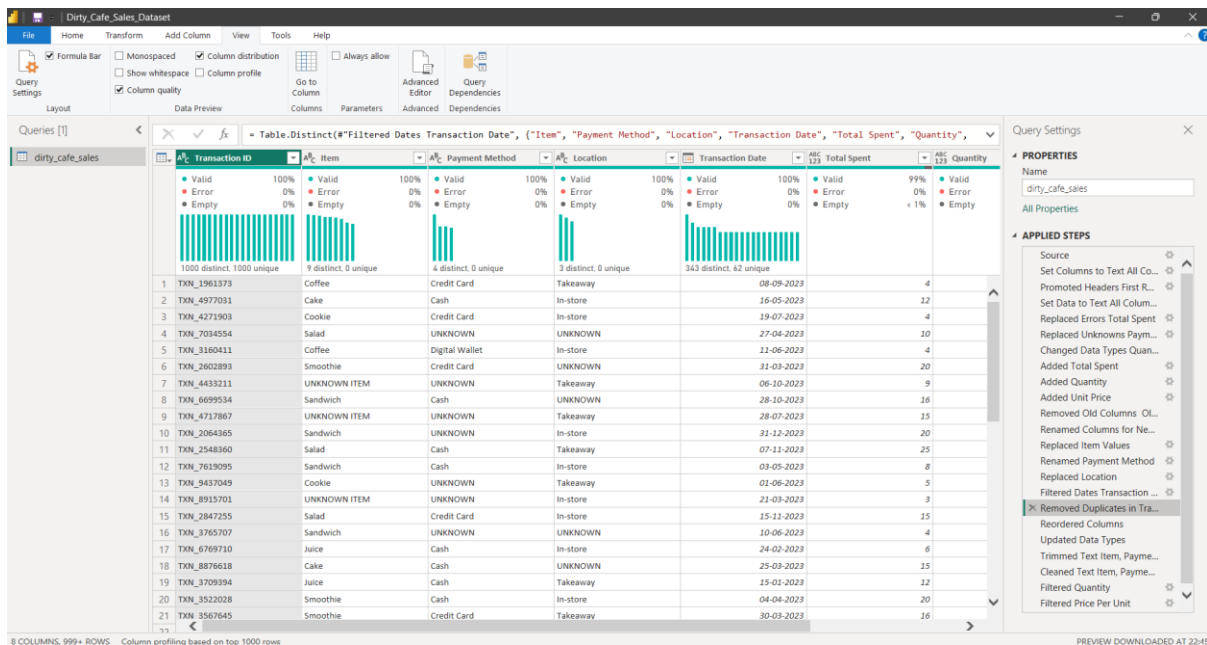


Fig. 19: Removed Duplicates in Transaction ID

18. Reordered Columns

The Reordered Columns step was used to arrange the fields into a logical order.

Although reordering columns does not change the underlying values, it improves the readability and usability of the dataset. Organizing related fields together makes it easier to understand the structure of the data and locate important information.

This step helped provide a cleaner and more organized layout for the final dataset.

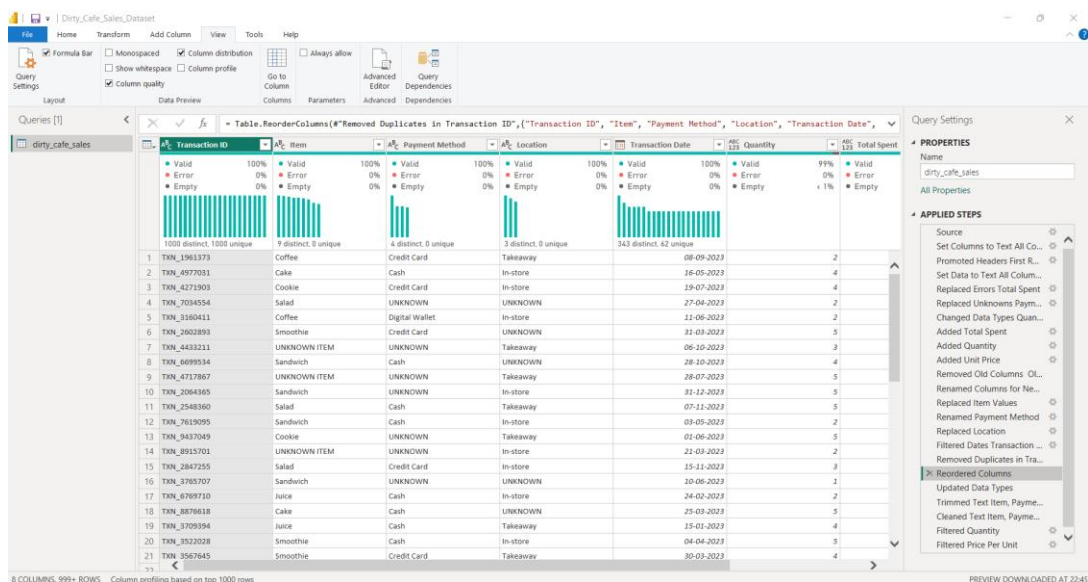


Fig. 20: Reordered Columns

19. Updated Data Types

The Updated Data Types step was applied after the main transformation operations to ensure that the required fields had suitable final data types.

Correct data types are essential for Power BI because numerical fields must be recognized as numbers and date fields must be recognized correctly for filtering and analysis. Text fields also need to remain appropriately formatted for categorical operations.

This step helped finalize the structure of the cleaned dataset and prepared it for further analytical use.

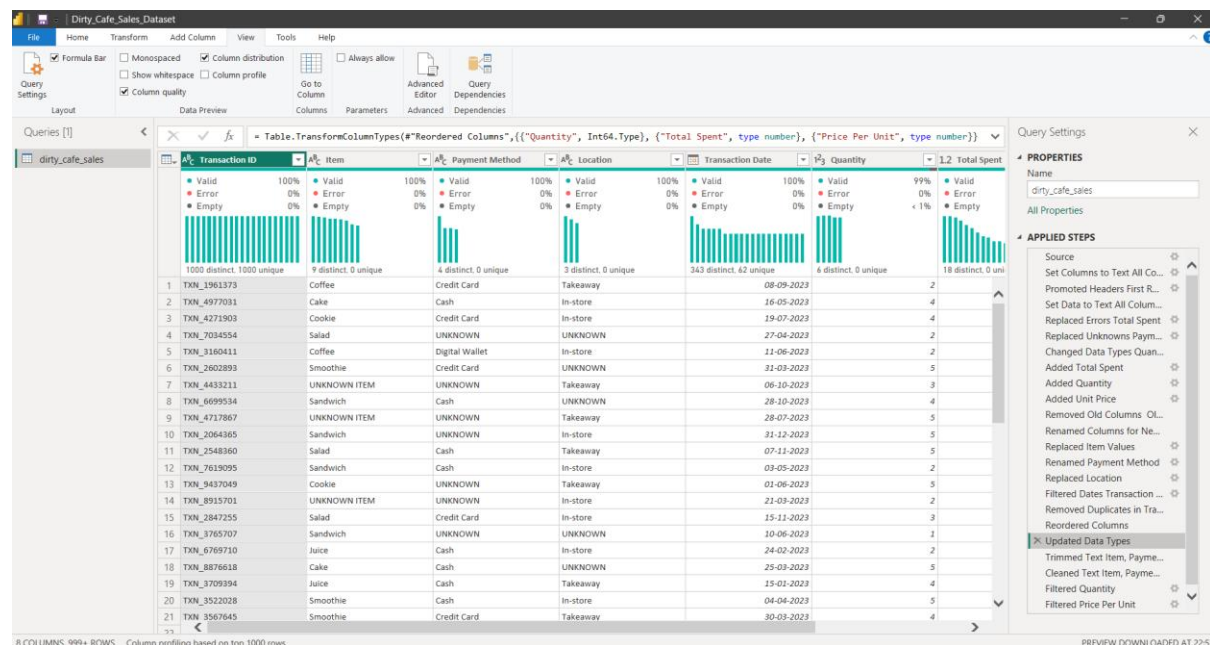


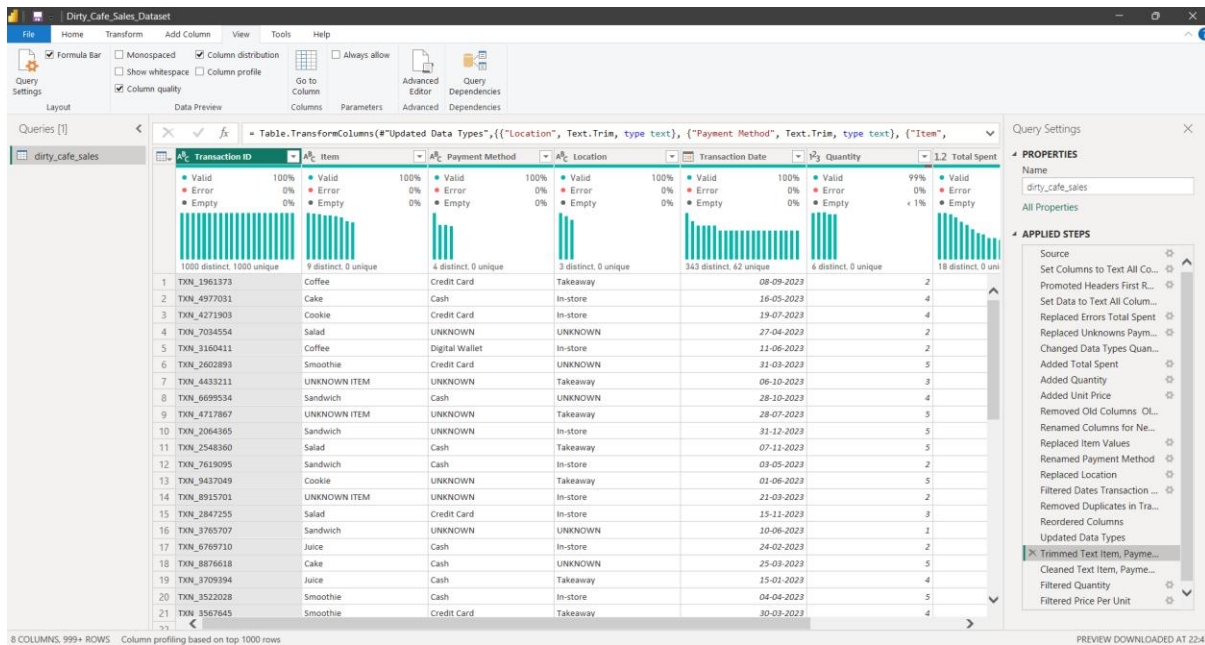
Fig. 21: Updated Data Types

20. Trimmed Text Item, Payment Method and Location

The Trimmed Text Item, Payment Method and Location step was applied to the relevant text fields to remove unnecessary spaces.

Extra spaces can cause values that appear identical to be treated as different values by the system. For example, a value containing an unwanted space may not match the same value without the space.

Trimming the selected text fields improved consistency and helped ensure that the categorical information was properly standardized.

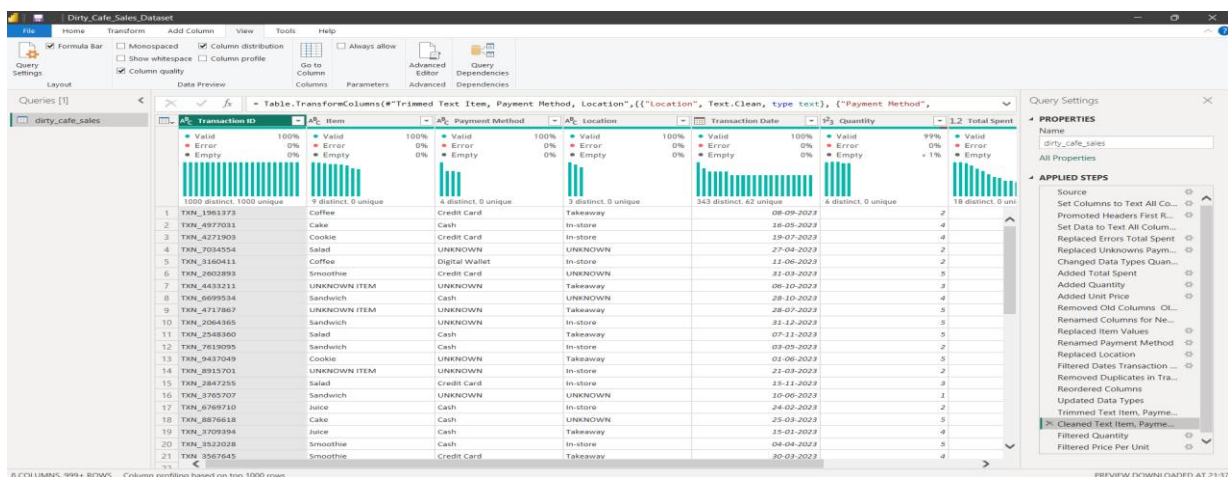


21. Cleaned Text Item, Payment Method and Location

The Cleaned Text Item, Payment Method and Location step was used to clean the relevant text fields after trimming.

Imported datasets can contain unwanted or non-printable characters that may not be immediately visible but can affect the quality of text-based analysis. The cleaning operation helped remove such unwanted characters from the selected fields.

Together with trimming, this step improved the overall quality and consistency of the text information.



22. Filtered Quantity

The Filtered Quantity step was applied to the Quantity field to retain suitable quantity values.

Since Quantity is a numerical field, invalid or unsuitable values can affect calculations and business analysis. Filtering the field helped ensure that the records carried forward into the final dataset satisfied the conditions applied during the cleaning process.

This improved the reliability of the Quantity information and prepared it for further analytical use.

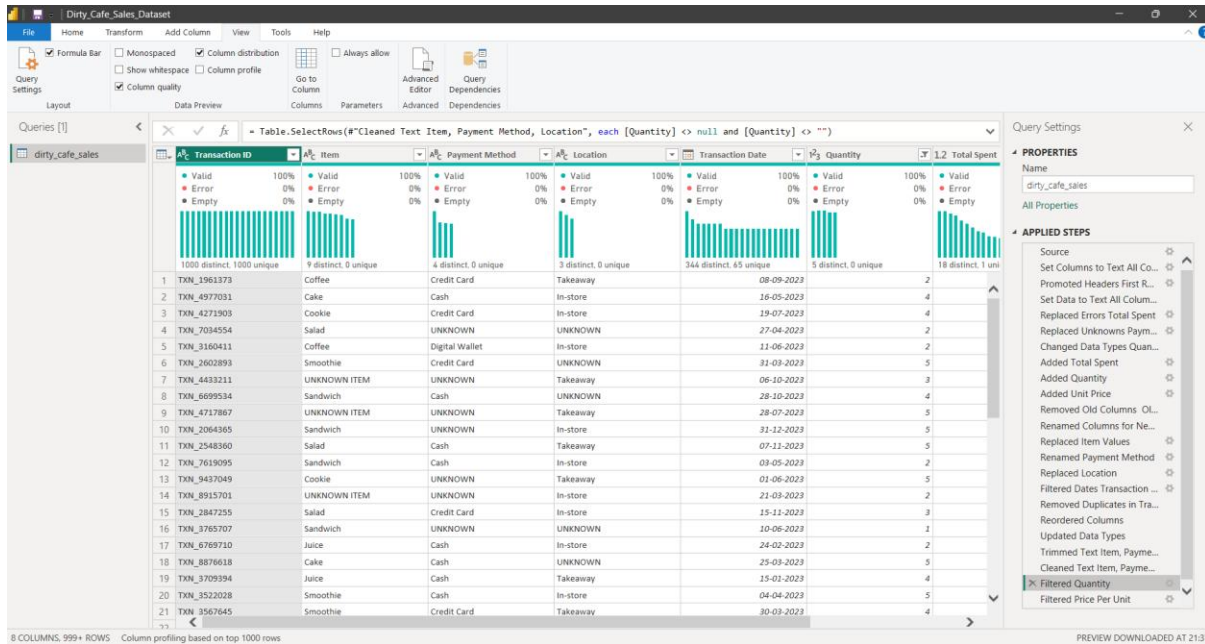


Fig. 24: Filtered Quantity

23. Filtered Price Per Unit

The Filtered Price Per Unit step was used to filter the Price Per Unit field.

Price Per Unit is an important numerical field because it represents the price associated with each unit of a purchased item. Unsuitable values can affect calculations and lead to inaccurate business conclusions.

The filtering operation helped retain the required price values and completed the main sequence of data-cleaning and validation operations performed in Power Query.

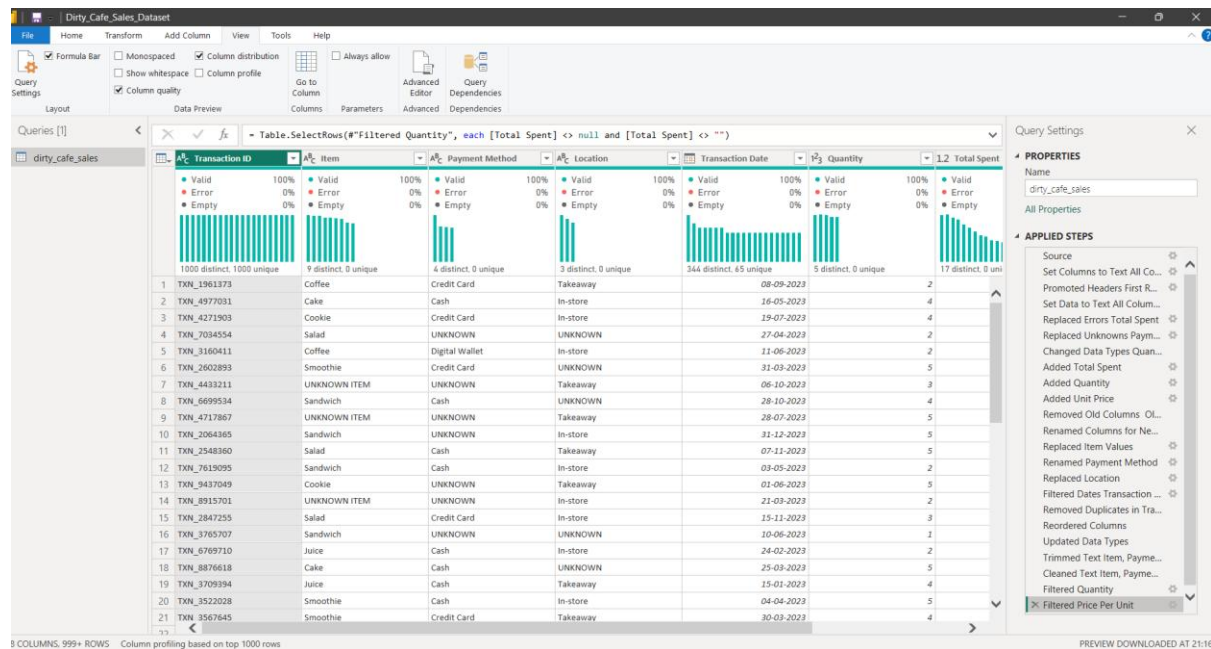


Fig. 25: Filtered Price Per Unit

Applied Steps Summary

The following table summarizes the major Applied Steps created during the Power Query cleaning process.

No.	Applied Step	Purpose
1	Imported Data	Imported the dataset into Power Query
2	Changed to Text	Set initial columns to text for preparation
3	Promoted Headers	Used the first row as column headers
4	Changed to Text 1	Applied text format where required
5	Replaced Errors	Handled error values
6	Replaced Unknown	Replaced unknown entries
7	Changed Data Types	Corrected required data types
8	Added Total Spent	Prepared Total Spent field
9	Added Quantity	Prepared Quantity field
10	Added Unit Price	Prepared Price Per Unit field
11	Removed Old Columns	Removed unnecessary old fields

12	Renamed Columns	Renamed prepared columns
13	Changed Data Types 1	Applied final numerical data types
14	Replaced Item Values	Standardized Item entries
15	Replaced Payment Method	Standardized Payment Method
16	Replaced Location	Standardized Location
17	Filtered Dates	Filtered transaction dates
18	Removed Duplicates	Removed duplicate transaction records
19	Trimmed Text	Removed unnecessary spaces
20	Cleaned Text	Removed unwanted characters
21	Filtered Quantity	Filtered invalid quantity values
22	Filtered Unit Price	Filtered invalid price values

The Applied Steps provide a clear transformation history of the dataset and make the cleaning process reproducible.

Step 4 – Final Data Validation and Preparation

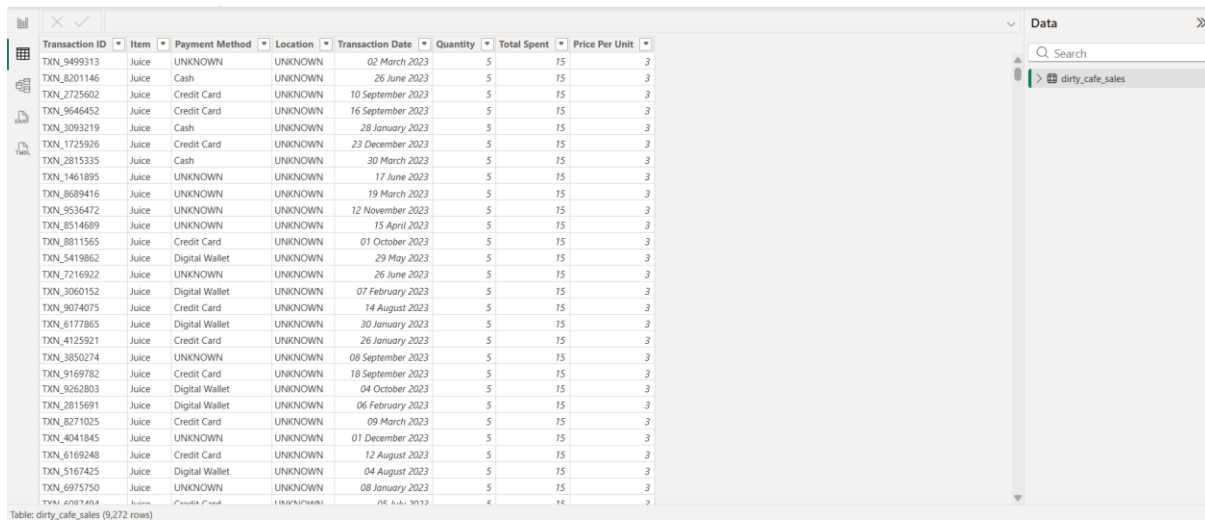
After applying all the required transformations, the final dataset was reviewed in Power Query.

The following checks were performed:

- Error values were handled.
- Unknown values were replaced.
- Required data types were corrected.
- Numerical fields were prepared.
- Unnecessary columns were removed.
- Column names were standardized.
- Item values were cleaned.
- Payment Method values were standardized.
- Location values were cleaned.
- Transaction dates were filtered.

- Duplicate transaction IDs were removed.
- Text values were trimmed.
- Text values were cleaned.
- Invalid quantity values were filtered.
- Invalid price values were filtered.

The final dataset was therefore prepared for reliable business analysis.



Transaction ID	Item	Payment Method	Location	Transaction Date	Quantity	Total Spent	Price Per Unit
TXN_9499313	Juice	UNKNOWN	UNKNOWN	02 March 2023	5	15	3
TXN_8201146	Juice	Cash	UNKNOWN	26 June 2023	5	15	3
TXN_2725602	Juice	Credit Card	UNKNOWN	10 September 2023	5	15	3
TXN_9646452	Juice	Credit Card	UNKNOWN	16 September 2023	5	15	3
TXN_3093219	Juice	Cash	UNKNOWN	28 January 2023	5	15	3
TXN_1725926	Juice	Credit Card	UNKNOWN	23 December 2023	5	15	3
TXN_2815335	Juice	Cash	UNKNOWN	30 March 2023	5	15	3
TXN_1461895	Juice	UNKNOWN	UNKNOWN	17 June 2023	5	15	3
TXN_8689416	Juice	UNKNOWN	UNKNOWN	19 March 2023	5	15	3
TXN_9536472	Juice	UNKNOWN	UNKNOWN	12 November 2023	5	15	3
TXN_8514689	Juice	UNKNOWN	UNKNOWN	15 April 2023	5	15	3
TXN_8811565	Juice	Credit Card	UNKNOWN	01 October 2023	5	15	3
TXN_5419862	Juice	Digital Wallet	UNKNOWN	29 May 2023	5	15	3
TXN_7216922	Juice	UNKNOWN	UNKNOWN	26 June 2023	5	15	3
TXN_3060152	Juice	Digital Wallet	UNKNOWN	07 February 2023	5	15	3
TXN_9074075	Juice	Credit Card	UNKNOWN	14 August 2023	5	15	3
TXN_6177865	Juice	Digital Wallet	UNKNOWN	30 January 2023	5	15	3
TXN_4125921	Juice	Credit Card	UNKNOWN	26 January 2023	5	15	3
TXN_3850274	Juice	UNKNOWN	UNKNOWN	08 September 2023	5	15	3
TXN_9169782	Juice	Credit Card	UNKNOWN	18 September 2023	5	15	3
TXN_9262803	Juice	Digital Wallet	UNKNOWN	04 October 2023	5	15	3
TXN_2015691	Juice	Digital Wallet	UNKNOWN	06 February 2023	5	15	3
TXN_8271025	Juice	Credit Card	UNKNOWN	09 March 2023	5	15	3
TXN_4041845	Juice	UNKNOWN	UNKNOWN	01 December 2023	5	15	3
TXN_6169248	Juice	Credit Card	UNKNOWN	12 August 2023	5	15	3
TXN_5167425	Juice	Digital Wallet	UNKNOWN	04 August 2023	5	15	3
TXN_6975750	Juice	UNKNOWN	UNKNOWN	08 January 2023	5	15	3
TXN_42072814	Juice	Credit Card	UNKNOWN	06 July 2023	5	15	3

Fig. 26: Final cleaned dataset in Power Query

Output

Before Cleaning

Before applying the Power Query transformations, the Dirty Cafe Sales Dataset contained several data quality issues.

These included:

- Error entries
- Unknown values
- Inconsistent text values
- Duplicate transactions
- Extra spaces
- Unwanted characters

- Invalid numerical values
- Data type inconsistencies

Preview file data

File path: C:\Users\Pragathi BR\OneDrive\Desktop\dirty_cafe_sales.csv

File origin: 65001: Unicode (UTF-8) Delimiter: Comma Data type detection: Based on first 200 rows

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8
Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card		2023-03-31
TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway	2023-10-06
TXN_6699534	Sandwich	4	4.0	16.0	Cash	UNKNOWN	2023-10-28
TXN_4717867		5	3.0	15.0		Takeaway	2023-07-28
TXN_2064365	Sandwich	5	4.0	20.0		In-store	2023-12-31
TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
TXN_3051279	Sandwich	2	4.0	8.0	Credit Card	Takeaway	ERROR
TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
TXN_9437049	Cookie	5	1.0	5.0		Takeaway	2023-06-01
TXN_8915701	ERROR	2	1.5	3.0		In-store	2023-03-21
TXN_2847255	Salad	3	5.0	15.0	Credit Card	In-store	2023-11-15
TXN_3765707	Sandwich	1	4.0	4.0			2023-06-10
TXN_6768710	Juice	2	3.0	6.0	Cash	In-store	2023-02-24
TXN_8876618	Cake	5	3.0	15.0	Cash	ERROR	2023-03-25

Load Transform data

Fig. 27: Dataset before cleaning

After Cleaning

After applying all the required Power Query transformations, the dataset was converted into a cleaner and more consistent form.

The final data contains standardized fields that are more suitable for analysis. The cleaning process also created a clear sequence of Applied Steps, allowing the transformations to be reviewed and reproduced.

Transaction ID	Item	Payment Method	Location	Transaction Date	Quantity	Total Spent	Price Per Unit
TXN_9499313	Juice	UNKNOWN	UNKNOWN	02 March 2023	5	15	3
TXN_8201146	Juice	Cash	UNKNOWN	26 June 2023	5	15	3
TXN_2725602	Juice	Credit Card	UNKNOWN	10 September 2023	5	15	3
TXN_9646452	Juice	Credit Card	UNKNOWN	16 September 2023	5	15	3
TXN_3093219	Juice	Cash	UNKNOWN	28 January 2023	5	15	3
TXN_1725926	Juice	Credit Card	UNKNOWN	23 December 2023	5	15	3
TXN_2815335	Juice	Cash	UNKNOWN	30 March 2023	5	15	3
TXN_1461895	Juice	UNKNOWN	UNKNOWN	17 June 2023	5	15	3
TXN_8689416	Juice	UNKNOWN	UNKNOWN	19 March 2023	5	15	3
TXN_9536472	Juice	UNKNOWN	UNKNOWN	12 November 2023	5	15	3
TXN_8514689	Juice	UNKNOWN	UNKNOWN	15 April 2023	5	15	3
TXN_8811565	Juice	Credit Card	UNKNOWN	01 October 2023	5	15	3
TXN_5419862	Juice	Digital Wallet	UNKNOWN	29 May 2023	5	15	3
TXN_7216922	Juice	UNKNOWN	UNKNOWN	26 June 2023	5	15	3
TXN_3060152	Juice	Digital Wallet	UNKNOWN	07 February 2023	5	15	3
TXN_9074075	Juice	Credit Card	UNKNOWN	14 August 2023	5	15	3
TXN_6177865	Juice	Digital Wallet	UNKNOWN	30 January 2023	5	15	3
TXN_4125921	Juice	Credit Card	UNKNOWN	26 January 2023	5	15	3
TXN_3850274	Juice	UNKNOWN	UNKNOWN	08 September 2023	5	15	3
TXN_9169782	Juice	Credit Card	UNKNOWN	18 September 2023	5	15	3
TXN_9262803	Juice	Digital Wallet	UNKNOWN	04 October 2023	5	15	3
TXN_2815691	Juice	Digital Wallet	UNKNOWN	06 February 2023	5	15	3
TXN_8271025	Juice	Credit Card	UNKNOWN	09 March 2023	5	15	3
TXN_4041945	Juice	UNKNOWN	UNKNOWN	01 December 2023	5	15	3
TXN_6169248	Juice	Credit Card	UNKNOWN	12 August 2023	5	15	3
TXN_5167425	Juice	Digital Wallet	UNKNOWN	04 August 2023	5	15	3
TXN_6975750	Juice	UNKNOWN	UNKNOWN	08 January 2023	5	15	3
TXN_6907404	Juice	Credit Card	UNKNOWN	05 July 2023	5	15	3

Table: dirty_cafe_sales (9,272 rows)

Fig. 28: Final cleaned dataset

Data Cleaning Summary

Data Quality Issue	Power Query Operation Used	Purpose
Error values	Replaced Errors	Handle invalid/error entries
Unknown values	Replaced Value	Standardize unknown entries
Incorrect data types	Changed Data Types	Prepare fields correctly
Unnecessary columns	Removed Columns	Simplify dataset
Inconsistent column names	Renamed Columns	Improve clarity
Incorrect item values	Replaced Item Values	Standardize item names
Payment method inconsistency	Replaced Payment Method	Standardize categories
Location inconsistency	Replaced Location	Standardize locations
Invalid dates	Filtered Dates	Retain suitable dates
Duplicate records	Removed Duplicates	Avoid repeated transactions
Extra spaces	Trimmed Text	Remove unnecessary spaces
Unwanted characters	Cleaned Text	Improve text quality
Invalid quantity	Filtered Quantity	Retain suitable quantity values
Invalid unit price	Filtered Unit Price	Retain suitable price values

Conclusion

The Dirty Cafe Sales Dataset was successfully cleaned and prepared using Power Query in Power BI. The experiment demonstrated how raw business data can contain errors, unknown values, duplicates, inconsistent text, incorrect data types, and invalid numerical records.

A systematic sequence of Power Query transformations was applied to address these issues. Error and unknown values were handled, data types were corrected, numerical fields were prepared, unnecessary columns were removed, column names were standardized, categorical fields were cleaned, dates were filtered, duplicate records were removed, and text values were trimmed and cleaned.

The final dataset is more consistent, organized, and suitable for further business analysis. The experiment also demonstrated the importance of data preparation as a fundamental step before creating meaningful reports, dashboards, and analytical models.

Learning Outcomes

After completing this experiment, the following learning outcomes were achieved:

- Learned how to import a business dataset into Power BI.
- Understood the purpose of Power Query in data preparation.
- Learned how to identify common data quality problems.
- Learned how to handle error and unknown values.
- Understood the importance of correct data types.
- Learned how to create and prepare numerical fields.
- Learned how to remove unnecessary columns.
- Learned how to rename and organize columns.
- Learned how to standardize categorical values.
- Learned how to filter invalid dates and numerical values.
- Learned how to remove duplicate transaction records.
- Learned how to trim and clean text values.
- Understood the importance of maintaining a sequence of Applied Steps.
- Learned how cleaned data can be prepared for further business analytics.

Overall, the experiment provided practical experience in transforming a raw and inconsistent business dataset into a cleaner and analysis-ready dataset using Power Query.